ABDULLAHI AYODEJI BELLO

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MOBILITY AI **GRAPH MACHINE LEARNING** **SPATIOTEMPORAL URBAN SYSTEMS**

PROFILE

PhD researcher in Civil and Environmental Engineering in Data & Systems track, focused on human mobility, urban systems, spatiotemporal modeling, graph machine learning, and simulation. Experienced working with city-scale mobility datasets, constructing transportation graphs, modeling station-level demand and imbalance, and translating spatial data into research, web applications, and decision-support tools.

EDUCATION

Northeastern University Boston, MA
PhD, Civil and Environmental Engineering – Data & Systems Track | GPA: 3.89/4.0 | Fully Funded | Expected May 2028
Coursework: Transportation Demand Forecasting & Modeling, Performance Modeling and Simulation of T. Networks

Northeastern University Boston, MA
MS, Urban Informatics | GPA: 3.91/4.0 | Full-Tuition Scholarship, CSSH Excellence Award | May 2025
Coursework: GIS; Advanced Spatial Analysis; Big Data for Cities; Computational Statistics; System Dynamics Modeling

Federal University of Technology Akure Nigeria
BTech, Urban and Regional Planning, with Honors | Dec 2019

TECHNICAL SKILLS

Machine Learning: Graph Neural Networks, spatiotemporal graph modeling, GRU, sequence modeling, trajectory prediction, demand forecasting, model evaluation
Programming & Data Science: Python, PyTorch, PyTorch Geometric, scikit-learn, pandas, NumPy, GeoPandas, NetworkX, SQL, R, Excel, Power BI
Mobility & Simulation: Human mobility modeling, bike-share systems, station-level imbalance prediction, origin-destination analysis, travel-time networks, demand simulation, time-window aggregation
Geospatial & Visualization: ArcGIS Pro, ArcGIS Online, Experience Builder, Web Maps, spatial joins, network analysis, raster/vector data, dashboards, data storytelling

RESEARCH & PROFESSIONAL EXPERIENCE

Ryan Wang Smart City Lab – Northeastern University Boston, MA
Research Assistant | Sep 2025 – Present
- Conduct big-data human mobility, transportation systems, and urban analytics research under PI supervision.
- Build spatiotemporal ML workflows for predicting bike-share station imbalance using trip activity, station capacity, weather, temporal features, and graph structure.
- Construct city-scale station graphs using origin-destination flows adjacency matrices for graph neural network experiments.
- Prototype GCN-GRU pipelines for forecasting station-level empty/full risk and imbalance ratios across 10-minute prediction windows.
- Design simulation logic to estimate natural demand patterns and system imbalance without operational intervention.

Boston Area Research Initiative (BARI) – Northeastern University Boston, MA
Research Assistant | Jul 2024 – Apr 2025; Sep 2025 – Present
- Analyze high-frequency environmental sensor data to study spatial variation in air quality, heat, and noise across Boston-area communities.
- Clean, aggregate, and evaluate sensor observations using Python-based exploratory and inferential analysis workflows.
- Create static and interactive maps, web apps, dashboards, and visualizations for community-facing reports and research presentations.
- Translate technical findings into accessible explanations for city partners, community organizations, and interdisciplinary teams.

Northeastern University – Planning, Real Estate and Facilities Boston, MA
GIS & Spatial Data Specialist | Jan 2024 – Jun 2024
- Developed, cleaned, modeled, and symbolized spatial datasets for campus infrastructure, including utilities, AEDs, CCTVs, buildings, amenities, and water features.
- Georeferenced, digitized, and updated 2D and 3D campus data to support facilities planning, infrastructure management, and campus mapping.
- Built GIS products including web applications, web maps, and static PDF maps for operational and planning use cases.

Multi Environmental Management Consultants – Lagos, Nigeria
Director of Operations, Environmental Research | Jun 2021 – Dec 2022
- Led research, data analysis, field coordination, and reporting for more than 30 environmental impact assessment projects.
- Integrated site observations, remote sensing, spatial analysis, and environmental planning methods to evaluate impacts and mitigation strategies.

Ministry of Physical Planning and Urban Development – Ogun, Nigeria
Urban Planner, National Service Corps | Mar 2020 – Mar 2021
- Operated drones to collect high-resolution aerial imagery and created vector/raster datasets for land-use planning and smart-city development work.

STARTUP AND IMPACT

BW Analytics Limited, Nigeria – *Founder & Managing Director* (Remote)
- Founded and lead a geospatial technology and urban analytics company developing data-driven tools for public-sector and societal challenges.
- Launched the Open Geospatial Data Initiative (OGDI) to curate and publish free, reusable geospatial datasets for research and public use, including datasets hosted on Harvard Dataverse.
- Established BW Analytics as a **Google Maps Content Partner**, contributing geospatial information to improve the availability and quality of location data.
- Led the development of 911 Rescue, a location-based emergency-response platform designed to improve access to police, hospitals, ambulance services, fire services, road-safety agencies, and other emergency providers.

Public Data Platforms & Events
- Common SENSES Data Platform **|** Platform for tracking real-time environmental conditions in historically disadvantaged communities; Developed using ESRI's experience builder **|** https://www.commonsensesproject.org/live-sensor-data
- Common SENSES Report Platform | This is an analytical report building web app that supports the data platform by performing ready-to-present analyses and insights to the public based on selected prompts | https://bells99k.github.io/CSENSES-Report-Builder/
- Harvard HSIL Hackathon in Lagos, Nigeria **|** Organized and sponsored Harvard University's Health Systems Innovation Lab global challenge for building health solutions leveraging AI | HSIL Website (under Lagos Nigeria) **|** BW Analytics LinkedIn post

SELECTED PROJECTS

Spatiotemporal Graph ML for Bike-Share Imbalance Prediction
PhD Research Project
- Built a station-level graph for Boston's bike-share system using trip flows, station capacity, travel-time relationships, weather, and neighborhood context.
- Modeled station imbalance as the ratio of minutes a station is empty or full within 10-minute time windows and engineered inflow/outflow, net-flow, rush-hour, weather, and temporal aggregation features.
- Developed baseline sequence models and graph-based GCN-GRU architecture to compare non-graph and graph-aware forecasting performance.

Sensor Bias Correction for Urban Environmental Monitoring
PhD Research Project
- Developed a bias-correction workflow for medium-cost environmental sensors by comparing sensor observations with reference-grade monitoring data and identifying systematic measurement errors.
- Processed high-frequency heat readings from air quality sensors across multiple monitoring locations, including timestamp alignment, missing-value handling, temporal aggregation, and quality-control checks.
- Evaluated corrected sensor outputs against reference observations to assess model performance and reduce bias in community-facing environmental monitoring results.

Urban Mobility Simulation for Natural Demand and System Failure Analysis
Research Prototype
- Simulated station inventory changes from observed trip traffic patterns to estimate natural system demand without rebalancing intervention and identify empty/full states, missed pickups/drop-offs, and recovery dynamics.

Geo-Deep Learning Change Detection in Urban Environments
Lead Author | Ongoing
- Developing research on deep-learning-based urban change detection using geospatial data and remote sensing methods.

PUBLICATIONS, CONFERENCES & PRESENTATIONS

- Conference Speaker, GIS-Pro 2026 – "Bike-share Performance and Prediction using GNN" Oct 2026.
- Conference Speaker, GIS-Pro 2024 – "Assessing Parking Demand Using Big Data and GIS," Oct 2024.
- Publication, Co-Author – "Building Damage Assessment in Aftermath of Disaster Events by Leveraging GeoAI," DOI: 10.30574/wjarr.2024.23.1.2000, Jul 2024.
- Presentation, City of Boston Analytics Team – "Assessing Parking Enforcement Incidents in Boston," Dec 2023.

CERTIFICATIONS & SERVICE

Certifications: 30+ Esri Academy certificates; Esri Spatial Data Science; SoloLearn SQL; SoloLearn Python; Nigerian German Center Data Science Certificate.
Memberships & Service: Geospatial Professional Network; American Planning Association; Esri Young Professionals Network; African Graduate Students Association - Program Director(2023-2024); Boston Debate League - Judge.
- Conference Proposal Reviewer – National Planning Conference 2026, Detroit; National Planning Conference 2025, Denver.
- Conrad Challenge 2025 Judge [Energy and Environment] – Presented by Equinox – Hosted by Space Center Houston.